

BETTER DATA. HEALTHIER TOMORROW.

HOSPITAL OPERATIONS & PATIENT ANALYTICS

Infosys Springboard Data Visualisation – Virtual Internship 7.0

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BETTER DATA
HEALTHIER
TOMORROW

INSIGHTS TODAY
FOR A HEALTHIER
TOMORROW

Care
Operate
Analyze
Improve



HOSPITAL OPERATIONS & PATIENT ANALYTICS

DATA-DRIVEN CARE
FOR A BRIGHTER TOMORROW



PATIENT
FLOW



BED
UTILIZATION



STAFF
EFFICIENCY



OPERATIONAL
INSIGHTS



BETTER
DECISIONS



People
Process
Progress
for a Healthier Tomorrow



Contents

From hospital data to operational insights



01
Abstract

03
Introduction

05
Proposed System Architecture

07
KPI Engineering

09
Tools & Technologies

02
Problem Statement & Objectives

04
Existing vs Proposed System

06
Dataset Resources

08
Results & Dashboard Analysis

10
Conclusion & Future Scope

Abstract

A concise view of the project and its value

Project Overview

Hospital operations generate large volumes of information across patients, admissions, departments, billing, beds, wards, and staff allocation. Analysing these sources separately can make operational trends difficult to identify. This project develops an interactive Hospital Operations and Patient Analytics Dashboard using Tableau. The workflow integrates hospital data, performs cleaning and transformation, calculates healthcare KPIs, and presents insights through four interconnected dashboards.

Data Integration

Combines related hospital tables into a unified analytical view.

KPI Monitoring

Converts operational data into clear, trackable measures.

Interactive Insights

Supports filtering, comparison and trend analysis.

Outcome

The dashboards help hospital administrators understand operational performance and support informed, data-driven decisions.

Problem Statement & Objectives

Why a centralised analytics view is needed

Problem Statement

Hospital information is often spread across admissions, billing, department and resource datasets. Separate analysis and manual reporting can make overall performance harder to understand, delay identification of trends, and limit timely operational decisions. An integrated dashboard is therefore needed to bring the information together and present clear, actionable insights.

Project Objectives

- Monitor admissions, discharges and patient flow.
- Compare department performance and efficiency.
- Track billing, revenue and key financial indicators.
- Evaluate bed, staff and ward resource utilisation.
- Support data-driven decisions through interactive dashboards.

Introduction

Connecting healthcare operations with data-driven decision-making

PATIENT CARE

OPERATIONS

ANALYTICS

DECISIONS



Why Analytics Matters

Hospitals coordinate patient registration, admissions, diagnosis, treatment, discharge, billing and resource allocation. These activities generate large volumes of operational data across multiple tables and files. When information is handled separately, reporting can become slower, and comparisons across departments can be difficult. Data integration and visualisation provide a single view of hospital performance, making important patterns easier to monitor and communicate.

Monitor

Track patient activity and operational trends.

Compare

Identify differences across departments and resources.

Understand

Translate complex datasets into readable visual insights.

Act

Use dashboard evidence to support operational decisions.

Existing System vs Proposed System

Moving from fragmented reporting to an integrated dashboard

Existing System

- Data maintained in separate files or tables.
- Manual analysis and reporting.
- Limited interactive visualisation.
- Difficult department-wise comparison.
- Operational trends may be identified late.



Proposed System

- Integrated hospital dataset.
- Cleaned and transformed data.
- Centralised KPI calculations.
- Interactive Tableau dashboards.
- Filters for detailed analysis.
- Unified dashboard navigation.
- Visual comparison of hospital operations.

Proposed System Architecture

End-to-end flow from hospital data to interactive dashboards

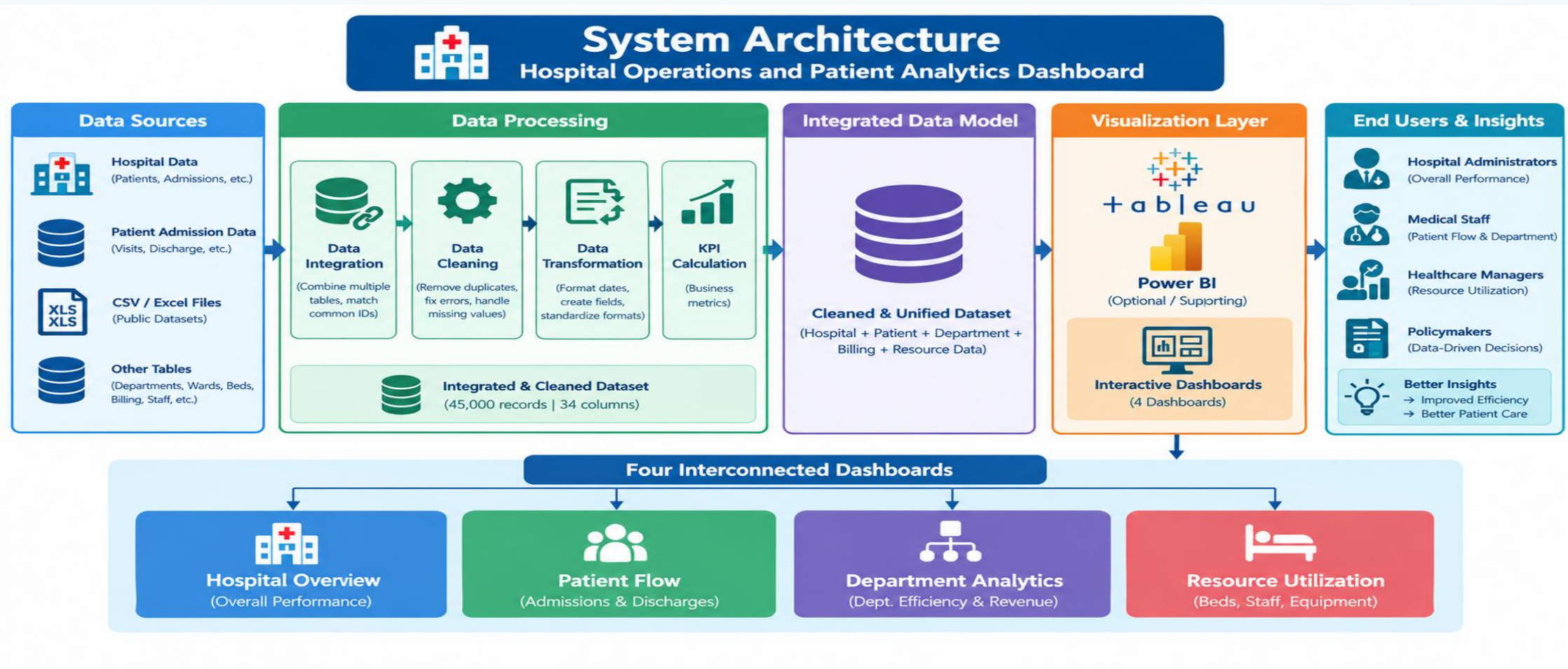


Fig1- System Architecture

Dataset Resources

Hospital HMIS data used for operational and patient analytics

Dataset

Source: Kaggle — Hospital HMIS Dataset for Healthcare Analytics

HMIS means **Hospital Management Information System**.

Dataset Role

This dataset is designed for healthcare analytics, dashboards and operational insights. It contains multiple interconnected tables. The **admission table** acts as the **central transactional table** which can be connected to other related tables using common keys.

Tables Used

Patient • Admission • Department • Ward • Bed • Disease • Billing • Employee • Staff Assignment

KPI Engineering

Turning hospital data into measurable performance indicators



Total Admissions

Counts unique Admission_ID values to indicate overall hospital workload.

Formula: Total Admissions =
COUNTD(Admission_ID)

Total Patients

Counts unique Patient_ID values to show the number of patients represented in the data.

Formula: Total Patients =
COUNTD(Patient_ID)

Total Discharges

Counts admissions with a recorded discharge date to help monitor patient flow.

Formula: Total Discharges =
COUNTD(Admission_ID), where
Discharge_Date is not null

Occupancy Rate

Measures how much of the available bed capacity is occupied over a period.

Formula: Occupancy Rate (%) = (Occupied Beds Days / (Total Beds*No of Days) × 100

Readmission Rate

Measures the proportion of unique patients who are readmitted.

Formula: Readmission Rate (%) = (total_readmitted_patients / total_unique_patients) * 100

KPI Engineering — Additional Measures

Operational metrics for stay duration, bed use and department performance

Average Length of Stay

Average number of days patients remain admitted. It supports analysis of patient flow, treatment duration and bed usage.

Formula: Average Length of Stay =
 $\text{SUM}(\text{Discharge_Date} - \text{Admission_Date})$

Bed Utilisation Rate

Measures how effectively available beds are used over the selected period.

Formula: Bed Utilisation Rate (%) =
 $(\text{occupied_beddays} / (\text{total_beds} * \text{no of days})) * 100$

Department Efficiency

A comparative measure used to understand department performance against the resources and operational output available to it.

Formula: Overall Department Efficiency (%) =
 $(\text{Dept Output} / \text{Dept Resources}) \times 100$

Why KPIs Matter

KPIs simplify large datasets into numbers that can be monitored quickly through dashboard cards. Together, they support trend analysis, comparison and identification of areas requiring attention.

Results & Dashboard Analysis

Four interconnected dashboards provide different operational views

Hospital Overview

Overall hospital performance, headline KPIs and financial/operational indicators.

Patient Flow

Admissions, discharges and patient activity trends.

Department Analytics

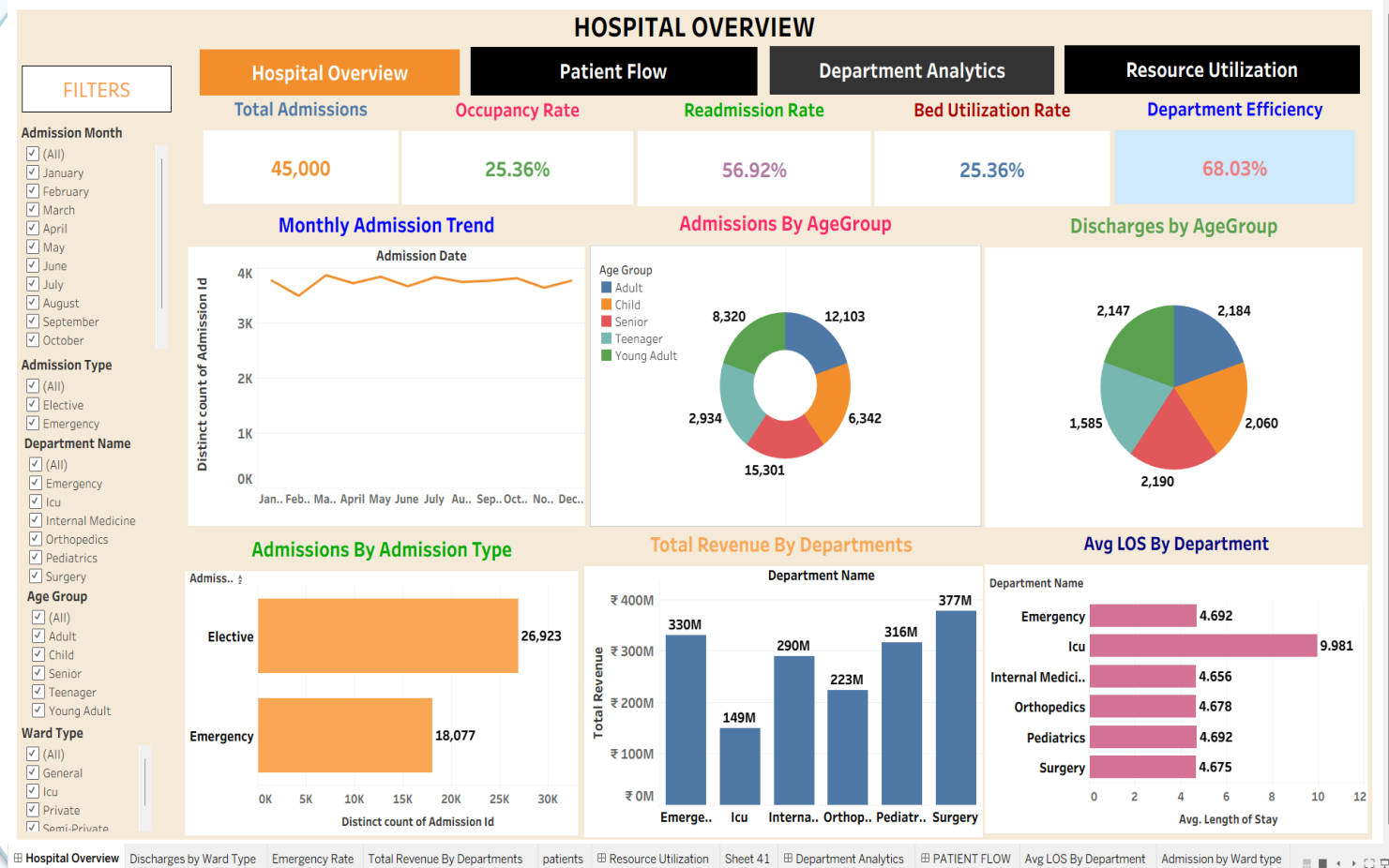
Department-level comparison, efficiency and revenue-related insights.

Resource Utilisation

Beds, staff and other resource utilisation indicators.

Hospital Overview Dashboard

Overall Performance

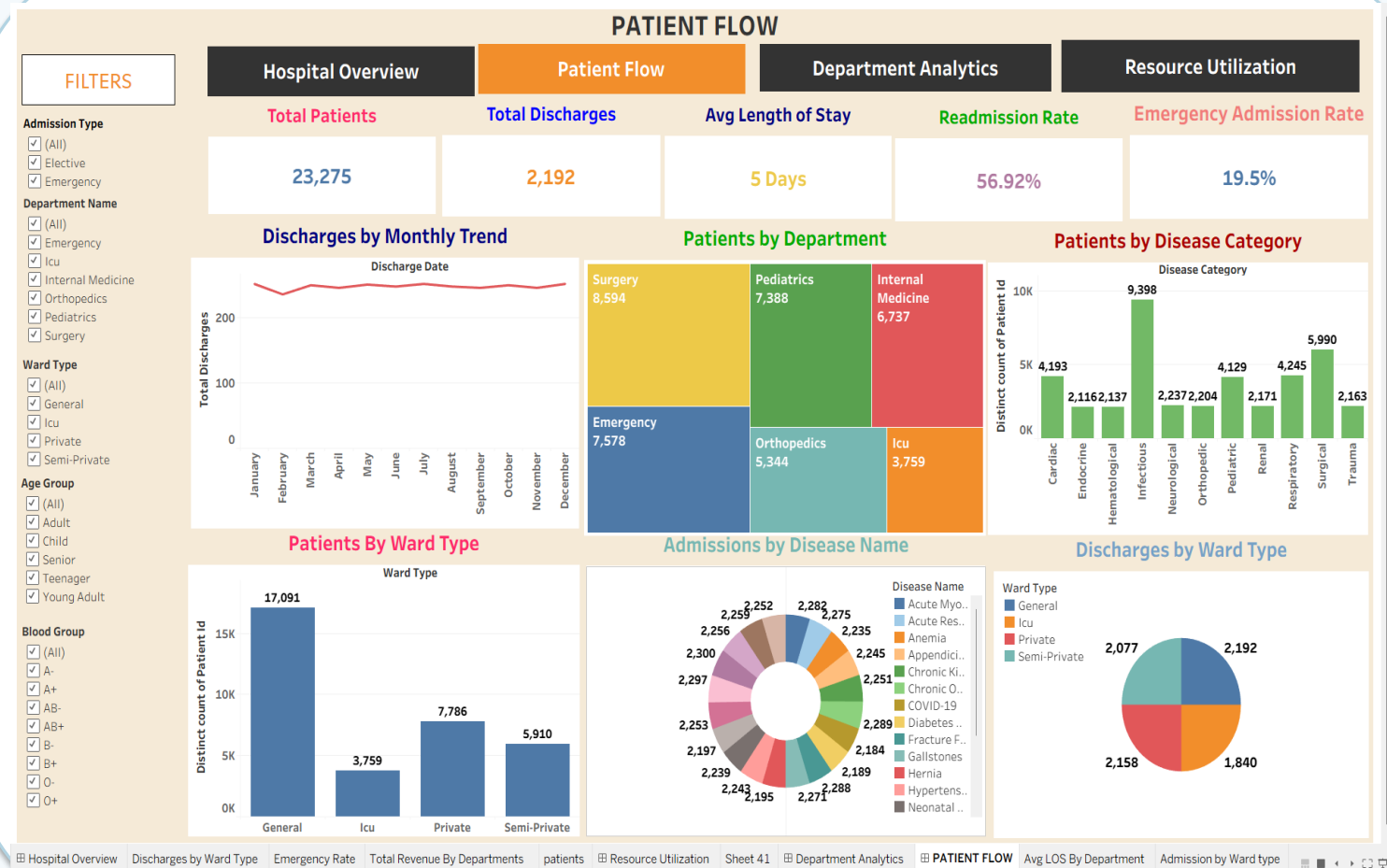


Key Takeaways

- Provides a consolidated view of hospital activity.
- Highlights key operational and financial indicators.
- Helps users understand overall performance at a glance.

Patient Flow Dashboard

Admissions & Discharges

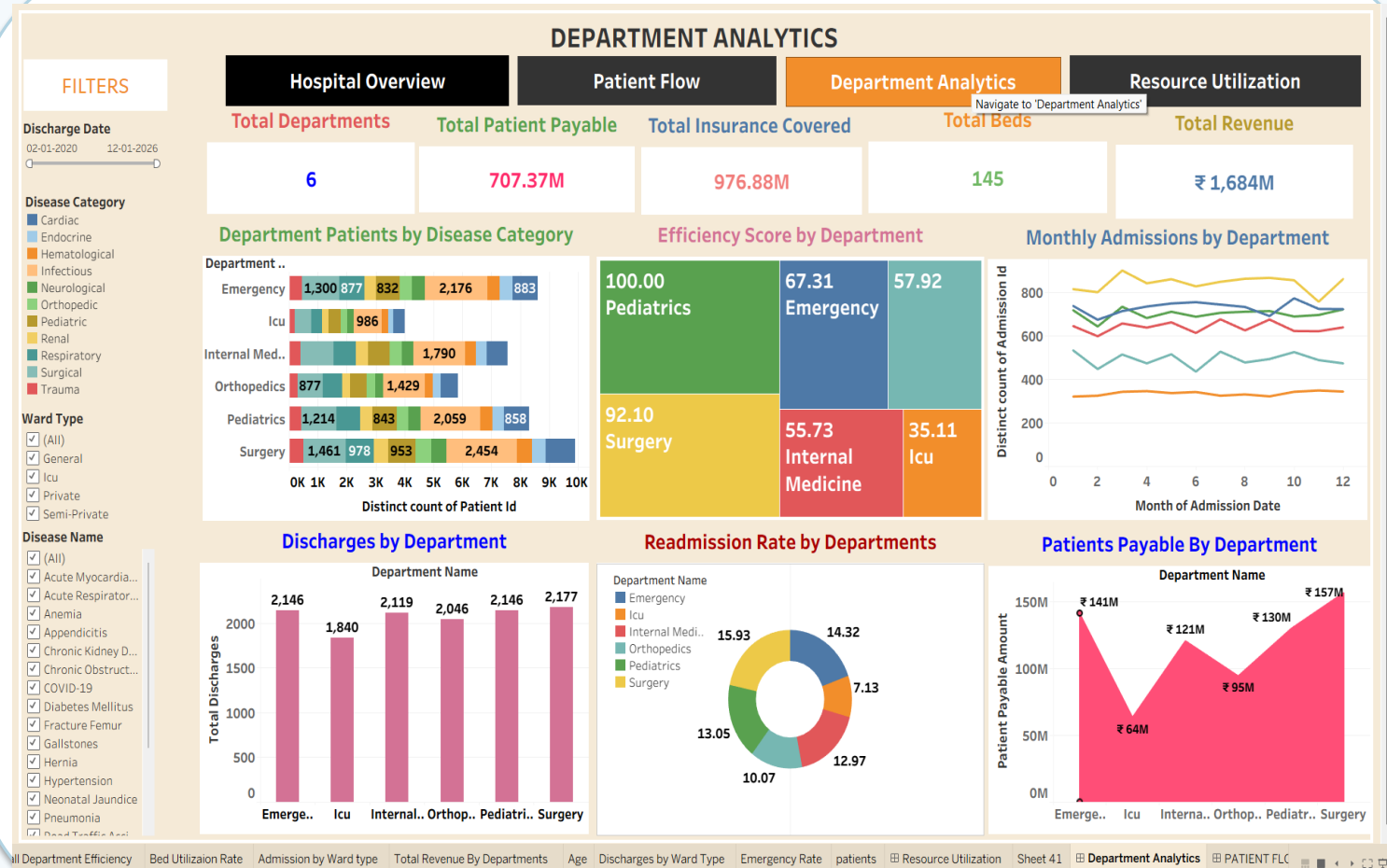


Key Takeaways

- Tracks patient movement through admissions and discharges.
- Supports trend analysis of patient activity.
- Helps identify changes in patient volume over time.

Department Analytics Dashboard

Department Performance



Key Takeaways

- Enables department-wise comparison.
- Supports analysis of performance, efficiency and revenue-related measures.
- Helps identify departments requiring closer operational attention.



Takeaways

FILTERS

Admission Type

☒ (All)
☒ Null
☒ Elective
☒ Emergency

Age Group

☒ Adult
☒ Child
☒ Senior
☒ Teenager
☒ Young Adult

Ward Type

☒ (All)
☒ General
☒ Icu
☒ Private
☒ Semi-Private

Department Name

☒ (All)
☒ Null
☒ Emergency
☒ Icu
☒ Internal Medicine
☒ Orthopedics
☒ Pediatrics

Disease Category

☒ (All)
☒ Cardiac
☒ Endocrine
☒ Hematological
☒ Infectious
☒ Neurological
☒ Orthopedic
☒ Pediatric
☒ Renal
☒ Respiratory
☒ Surgical

Hospital Overview

Total Beds

145

Patient Flow

Total Staff

70

Department Analytics

Total Wards

27

Resource Utilization

Avg Beds per Ward

5

Bed Utilization Rate

25.36%

Discharges by Disease Category

Disease Category	Total Discharges
Car.	1,900
En.	1,374
He.	1,401
Inf.	2,186
Ne.	1,434
Ort.	1,431
Pe.	1,907
Re.	1,385
Res.	1,929
Sur..	2,074
Tra.	1,422

Total Beds by Department

Department	Beds
Surgery	33
Pediatrics	27
Internal Medicine	25
Emergency	28
Orthopedics	19
Icu	13

Beds by Ward Type

Ward Type	Beds
General	82
Icu	13
Private	29
Semi-Private	21

Avg Staff Allocation by Department

Department	Staff Count
Emergency	11
Icu	12
Internal Medicine	10
Orthopedics	14
Pediatrics	11
Surgery	12

Bed Utilization by Department

Month	Emergency (%)	Icu (%)	Internal Med. (%)	Orthopedics (%)	Pediatrics (%)	Surgery (%)
Jan.	1.4	1.4	2.8	2.3	3.2	3.4
Feb.	1.4	1.4	2.7	2.2	3.1	3.5
Mar.	1.4	1.4	2.8	2.3	3.2	3.4
Apr.	1.4	1.4	2.7	2.2	3.1	3.5
May	1.4	1.4	2.8	2.3	3.2	3.4
Jun.	1.4	1.4	2.7	2.2	3.1	3.5
July	1.4	1.4	2.8	2.3	3.2	3.4
Aug.	1.4	1.4	2.7	2.2	3.1	3.5
Se.	1.4	1.4	2.8	2.3	3.2	3.4
Oc.	1.4	1.4	2.7	2.2	3.1	3.5
No.	1.4	1.4	2.8	2.3	3.2	3.4
De.	1.4	1.4	2.7	2.2	3.1	3.5

Department Wise Payable By Age Group

Department	Emergent (₹M)	Critical (₹M)	Intermediate (₹M)	Orthopedic (₹M)	Pediatric (₹M)	Surgical (₹M)
Emergency	26	21	48	18	38	39
Icu	23	17	40	18	32	23
Internal	23	30	45	18	37	55
Orthop.	23	18	30	45	37	23
Pediatr.	23	23	45	18	37	39
Surgery	29	23	55	18	37	39

Department Efficiency | Bed Utilization Rate | Admission by Ward type | Total Revenue By Departments | Age | Discharges by Ward Type | Emergency Rate | patients | Resource Utilization | Sheet 41 | Department Analytics | PATIENT FLC

- Monitors utilisation of hospital resources.
- Supports bed and staffing analysis.
- Helps management understand capacity and resource usage.

Tools & Technologies

What each tool contributes to the project

Kaggle

Dataset Source

Provides the Hospital HMIS dataset used as the project's analytical data source.

Tableau

Dashboard Visualisation

Primary platform used to build the four interactive hospital dashboards and present KPIs.

GitHub

Version Control

Helps maintain project files, track changes and support collaborative development.

Python

Data Processing

Supports data preparation activities such as cleaning, transformation and analytical processing.

Power BI

Supporting Visualisation

Included as a supporting/alternative visualisation platform in the system architecture.

Technology Flow

Dataset → Data Preparation → KPI Engineering → Tableau Dashboards → Operational Insights

Conclusion

From fragmented hospital data to a unified analytical view

Project Conclusion

The Hospital Operations and Patient Analytics Dashboard integrates hospital operational data and converts it into meaningful healthcare insights. The solution brings together KPI monitoring, trend analysis, department comparison, financial indicators and resource utilisation in a centralised visual environment. This project demonstrates how data analytics and visualisation can support hospitals in understanding operational performance and making informed decisions.

Hospital Overview

Overall performance and key indicators.

Patient Flow

Admissions and discharge trends.

Department Analytics

Department comparison and performance.

Resource Utilisation

Beds, staff and resource usage.

Future Scope

Potential extensions for a more advanced hospital analytics solution



Real-Time Monitoring

Connect live hospital systems so KPIs and resource status can update continuously.

Predictive Analytics

Forecast patient volumes, bed demand and operational trends using historical data.

Early-Warning Indicators

Create alerts for unusual changes in occupancy, readmissions or patient flow.

Advanced Resource Planning

Use analytics to support staffing, bed allocation and capacity planning.

Expanded Data Integration

Integrate additional hospital sources to create a broader operational view.

Role-Based Dashboards

Provide customised views for administrators, medical staff and healthcare managers.

THANK YOU

Hospital Operations & Patient Analytics Dashboard

Vinjam Sindhura

